📌 Athlete Details

- **Name:** Rahul Verma
- **Age:** 23
- **Sport:** Athletics (Sprinter)
- **Athlete ID:** ATH-1023
- **Monitoring Type:** IoT Wearable Sensors

📡 IoT Sensor Data (Last 14 Days)

Parameter	Baseline	Current	Change
Heart Rate (BPM)	62	48	↓ 22%
Hematocrit (%)	44	52	↑ 18%
Testosterone (ng/dL)	520	890	↑ 71%
Oxygen Saturation (%)	98	95	↓ 3%
Recovery Time (hrs)	18	28	↑ 55%
VO2 Max	55	60	↑ 9%

📈 Time-Series Observation

- Sudden spike in **Testosterone levels** over 10 days
- Significant increase in **Hematocrit levels**
- Decrease in **Heart Rate (possible artificial enhancement)**
- Increase in recovery time → possible physiological stress

🚨 Detected Anomalies

- Abnormal increase in Testosterone (+71%)
- Hematocrit above safe athletic threshold (>50%)
- Combined low heart rate + high RBC indicators
- Reduced oxygen efficiency

🧠 AI Medical Analysis

Condition Summary:
The athlete shows multiple abnormal physiological changes when compared to their baseline data.

Abnormal Findings:

- Elevated hematocrit indicates increased red blood cell concentration
- Testosterone spike is not consistent with natural variation
- Decreased heart rate combined with increased oxygen demand is unusual

Possible Causes:

- Performance Enhancing Drug (PED) usage (EPO or anabolic steroids)
- Artificial blood doping
- Overtraining or recovery imbalance

🕵️ Doping Suspicion Indicator

🔴 **Possible Doping Pattern Detected**

🎯 Risk Score

Risk Level: 78 / 100

👉 **High Risk**

⚠️ Confidence Score

Confidence: 84%

(Based on strong anomaly patterns + consistent deviations)

📊 Comparison with Normal Athlete

Metric	Normal Range	Athlete Value	Status
Hematocrit	38–48%	52%	❌ High
Testosterone	300–700	890	❌ High
Heart Rate	55–75 BPM	48	⚠️ Low

🚨 System Alerts

- ⚠️ Sudden hormonal spike detected
- ⚠️ Blood parameter imbalance
- 🚨 Possible doping-related pattern

🏁 Final Decision

🚫 **HIGH RISK FLAG**

👉 Athlete is **NOT CLEARED to compete**

👉 Immediate medical investigation recommended

📌 Recommended Actions

- Conduct anti-doping lab tests
- Monitor biomarkers daily
- Restrict from competitive events temporarily
- Perform full physiological evaluation

🔥 How to use this in your project

✅ For demo:

- Paste this as sample report
- Show AI analysis output

✅ For graphs:

Use values:

```
Heart Rate: [62, 60, 58, 55, 52, 50, 48]
```

Testosterone: [520, 540, 580, 650, 720, 800, 890]
Hematocrit: [44, 45, 46, 48, 49, 50, 52]